

Brain Tumor Detection Using Convolutional Neural Networks (CNN)

Training Project Report

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Project Title

Brain Tumor Detection Using Convolutional Neural Networks (CNN)

Tools & Technologies

- Python
- TensorFlow
- Keras
- NumPy
- Matplotlib
- OpenCV
- Visual Studio Code
- GitHub

Acknowledgement

I would like to express my sincere gratitude to **Ma'am Gulshan Melvi** for her continuous guidance, encouragement, and valuable knowledge throughout the Sherozgar AI & Machine Learning Training Program. Her practical teaching approach enabled me to understand the fundamentals of Machine Learning and Deep Learning and apply them in a real-world project.

I am also thankful to the Sherozgar Training Program for providing an excellent learning environment and practical exposure to Artificial Intelligence technologies.

Abstract

This project was completed as the final practical assignment of the **Sherozgar AI & Machine Learning Training Program**. The objective was to apply the concepts learned during the training by developing a real-world Artificial Intelligence application.

The project implements a **Convolutional Neural Network (CNN)** to classify brain MRI images into two categories: **Brain Tumor** and **No Brain Tumor**. The system was developed using Python, TensorFlow, and Keras. MRI images were preprocessed, used to train the CNN model, and evaluated on unseen testing data.

The trained model achieved a **95.09% testing accuracy**, demonstrating the effectiveness of CNNs for medical image classification and providing practical experience in Deep Learning model development.

1. Introduction

Artificial Intelligence (AI) is transforming various industries, particularly healthcare. One of its most valuable applications is medical image analysis, where Deep Learning models can assist healthcare professionals in identifying diseases from medical scans.

During the Sherozgar AI & Machine Learning Training Program, several concepts were studied, including supervised learning, unsupervised learning, clustering algorithms, Artificial Neural Networks (ANNs), and Convolutional Neural Networks (CNNs). This project applies those concepts by building a CNN model capable of detecting brain tumors from MRI images.

The project demonstrates the complete machine learning workflow, including data preprocessing, model development, training, evaluation, prediction, and GitHub project management.

2. Problem Statement

Manual analysis of MRI scans requires experienced medical professionals and can be time-consuming. This project addresses the following problem:

"Can a Convolutional Neural Network accurately detect the presence of a brain tumor from MRI images?"

The goal is to build an AI model that classifies MRI images into Tumor and No Tumor categories with high accuracy.

3. Objectives

The objectives of this project are:

- Apply Deep Learning concepts learned during training.
- Build a CNN model using TensorFlow and Keras.
- Train the model on MRI brain images.
- Classify MRI scans into Tumor and No Tumor.
- Evaluate model performance.
- Save the trained model for future predictions.
- Upload the complete project to GitHub.

4. Dataset

The Brain Tumor MRI Dataset was used for this project.

The dataset contains two classes:

- Yes (Brain Tumor)
- No (Healthy Brain)

The dataset was divided into:

- Training Dataset
- Testing Dataset

Dataset Statistics:

- Training Images: **7012**
- Testing Images: **1752**

Before training, all images were resized to **224 × 224 pixels** and normalized to improve model performance.

5. Methodology

The project followed these development stages:

Step 1 – Data Collection

The MRI dataset was downloaded and organized into training and testing folders.

Step 2 – Data Preprocessing

ImageDataGenerator was used to normalize images and prepare batches for training.

Step 3 – CNN Model Development

The CNN architecture included:

- Convolution Layer
- Max Pooling Layer
- Convolution Layer
- Max Pooling Layer

- Convolution Layer
- Max Pooling Layer
- Flatten Layer
- Dense Layer
- Dropout Layer
- Output Layer (Sigmoid Activation)

The model was compiled using:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Evaluation Metric: Accuracy

Step 4 – Model Training

The model was trained for **5 epochs** using the training dataset while validation was performed using the testing dataset.

Step 5 – Model Evaluation

The trained model was evaluated using unseen testing images to measure its prediction performance.

Step 6 – Prediction

A separate prediction script (**predict.py**) was created to classify new MRI images using the saved CNN model.

6. Results

The project successfully classified MRI brain images.

Final Results:

- Training Images: **7012**
- Testing Images: **1752**
- Epochs: **5**
- Test Accuracy: **95.09%**
- Test Loss: **0.1407**

The trained model was successfully saved as:

brain_tumor_model.keras

Training accuracy and loss graphs were generated and stored in the project results folder.

7. Project Structure

The project was organized professionally using the following structure:

- dataset/

- models/
- results/
- report/
- train.py
- predict.py
- README.md
- requirements.txt

The project was uploaded to GitHub as required by the training program.

8. Challenges Faced

Several practical challenges were encountered during development, including:

- Installing TensorFlow and required Python libraries.
- Managing Python virtual environments.
- Organizing the dataset correctly.
- Resolving file path errors.
- Running Deep Learning training on a CPU-only system.
- Saving and loading the trained CNN model.

These challenges improved my practical problem-solving skills and strengthened my understanding of AI project development.

9. Learning Outcomes

Through this project, I gained practical experience in:

- Python programming
- Image preprocessing
- Convolutional Neural Networks
- TensorFlow and Keras
- Model training and evaluation
- GitHub project management
- Deep Learning workflow
- Medical image classification

This project strengthened my confidence in applying AI concepts to solve real world problems.

10. Future Improvements

The project can be further enhanced by:

- Using larger MRI datasets.
- Applying data augmentation techniques.

- Implementing transfer learning models such as ResNet50 or EfficientNet.
- Developing a web-based application.
- Supporting multi-class brain tumor classification.
- Deploying the model for real-time prediction.

11. Conclusion

This project successfully fulfilled the objectives of the Sherozgar AI & Machine Learning Training Program by implementing a complete Deep Learning solution for brain tumor detection. The CNN model achieved a testing accuracy of **95.09%**, demonstrating its ability to classify MRI images effectively.

Beyond the technical implementation, this project provided valuable practical experience in dataset preparation, CNN model development, training, evaluation, prediction, and project management using GitHub. It strengthened my understanding of Artificial Intelligence and Deep Learning and motivated me to continue learning advanced AI techniques in the future.

References

1. TensorFlow Documentation
2. Keras Documentation
3. Brain Tumor MRI Dataset (Kaggle)
4. Goodfellow, I., Bengio, Y., & Courville, A. *Deep Learning*. MIT Press.
5. Géron, A. *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*. O'Reilly Media.